

RAPPORT DE TRAVAUX PRATIQUES :

CYBERSÉCURITÉ TP-1

1. Introduction

Ce rapport présente les résultats des manipulations effectuées dans un environnement de laboratoire contrôlé (VirtualBox). L'objectif est d'adopter la posture d'un auditeur en sécurité pour cartographier un réseau, identifier les services exposés et analyser les vulnérabilités potentielles d'une cible.

- Environnement de test :
 - Machine Attaquante : Kali Linux (192.168.56.30)
 - Cible Linux : Ubuntu Server (192.168.56.10)

2. TP 1.1 — Découverte et Cartographie Réseau (Nmap)

2.1.PRESENTATION DE Nmap

Nmap est un outil de scan réseau indispensable en cybersecuriter dont le role principale est de cartographier un reseau et didentifier ses services.

Le context d'utilisation de Nmap s'inscrit principalement dans les phases initiales dun audit de sécurité ou dune intrusion. Il sert de pont entre la decouverte brute dun reseau de l'exploitation de failles specifiques.

2.2. ANALYSE DES COMMANDES UTILISEES

-sn desactive le scan de port apres decouverte

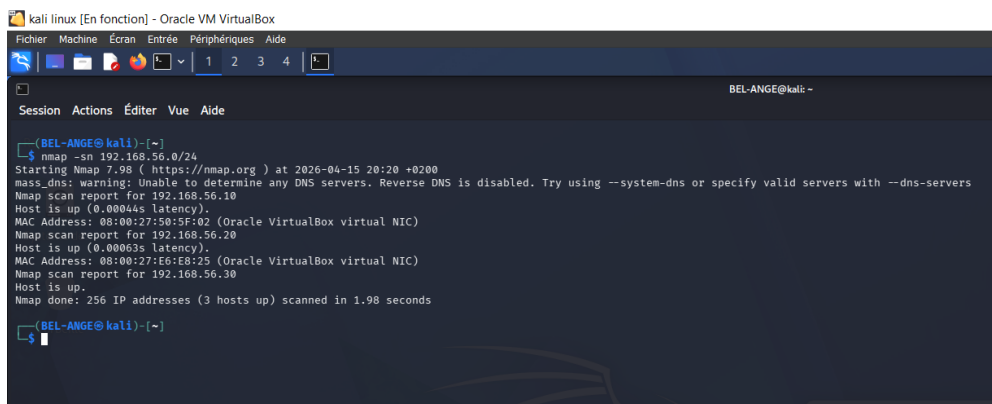
-Ss cest le scan par defaultn et le plus populaire

-Sv interroge les ports ouverts pour determiner exactement quell service tourne et quel est sa version précise

-o analyse les réponses des paquets IP/TCP pour deviner le systeme dexploitation de la cible.

2-3) RESULTAT OBTENUS

Commande: **nmap -sn 192.168.56.0/24**



```
kali linux [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
1 2 3 4
BEL-ANGE@kali ~
Session Actions Éditer Vue Aide
(BEL-ANGE@kali)~$ nmap -sn 192.168.56.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 20:20 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.00044s latency).
MAC Address: 08:00:27:50:5F:02 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.20
Host is up (0.00063s latency).
MAC Address: 08:00:27:E6:E8:25 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.30
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 1.98 seconds
(BEL-ANGE@kali)~$
```

Commande: **sudo nmap -sS,-sV,-O 192.168.56.10**

PORTS ET SERVICES DÉTECTÉS

```
kali linux [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
BEL-ANGE@kali -
Session Actions Éditer Vue Aide
(BEL-ANGE@kali)~$
$ nmap -sS 192.168.56.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 20:21 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.0002s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:50:5F:02 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds
(BEL-ANGE@kali)~$
```

```
kali linux [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
BEL-ANGE@kali -
Session Actions Éditer Vue Aide
(BEL-ANGE@kali)~$
$ nmap -sV 192.168.56.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 20:22 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.0000s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.9p1 Ubuntu 3ubuntu3 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache/2.4.63 ((Ubuntu))
MAC Address: 08:00:27:50:5F:02 (Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.11 seconds
(BEL-ANGE@kali)~$
```

```
kali linux [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
BEL-ANGE@kali -
Session Actions Éditer Vue Aide
(BEL-ANGE@kali)~$
$ nmap -O 192.168.56.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 20:23 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.0010s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:50:5F:02 (Oracle VirtualBox virtual NIC)
Device type: general purpose/router
Running: Linux 4.X|5.X, Mikrotik RouterOS 7.X
OS CPEs: cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), Mikrotik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.87 seconds
(BEL-ANGE@kali)~$
```

2-4) ANALYSE DE SECURITER

*Services dangereux

-apache (port 80)

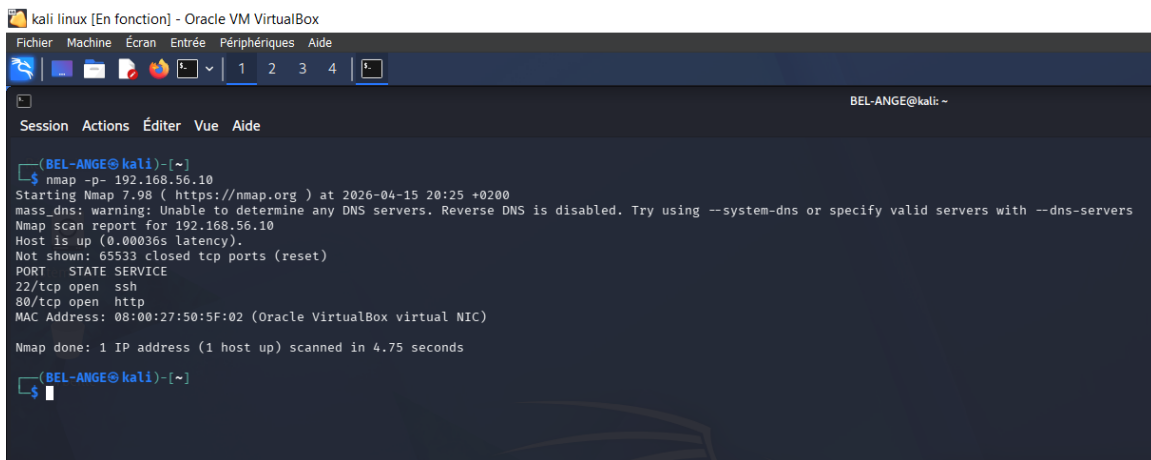
-openSSH (port 22)

*ports a fermer: port 80 (http), ports non identifier

*risqué pour l'entreprise: la fuite d'information, la compromission totale, le déni de service

2-5) utilisation avancée de Nmap

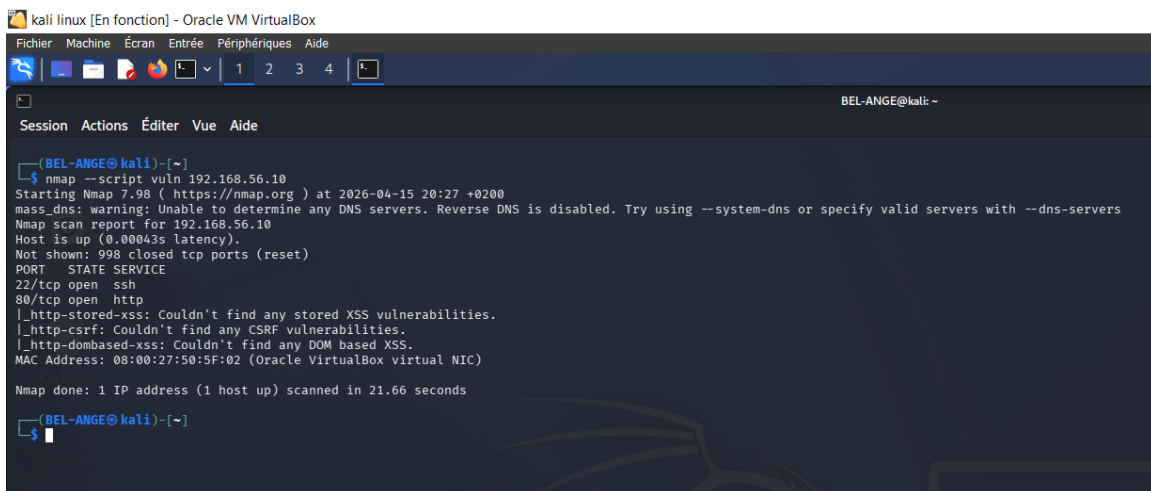
-p-



```
kali linux [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
1 2 3 4
BEL-ANGE@kali: ~
Session Actions Éditer Vue Aide
(BEL-ANGE@kali)-[~]
$ nmap -p- 192.168.56.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 20:25 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.00036s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:50:5F:02 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.75 seconds
(BEL-ANGE@kali)-[~]
$
```

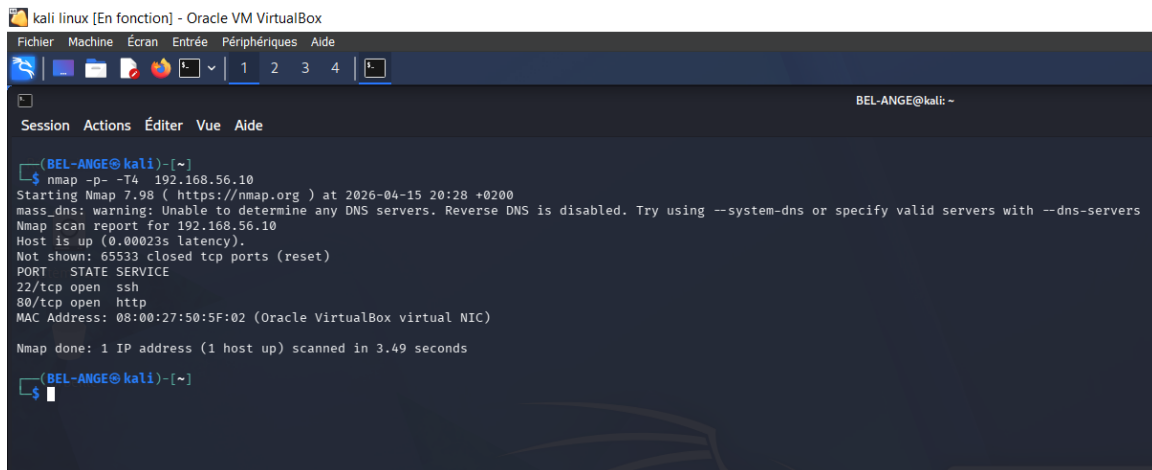
--script vuln



```
kali linux [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
1 2 3 4
BEL-ANGE@kali: ~
Session Actions Éditer Vue Aide
(BEL-ANGE@kali)-[~]
$ nmap --script vuln 192.168.56.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 20:27 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.00043s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
MAC Address: 08:00:27:50:5F:02 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 21.66 seconds
(BEL-ANGE@kali)-[~]
$
```

-T4



```
kali linux [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
1 2 3 4
BEL-ANGE@kali: ~
Session Actions Éditer Vue Aide
(BEL-ANGE@kali)-[~]
$ nmap -p- -T4 192.168.56.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 20:28 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.00023s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:50:5F:02 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 3.49 seconds
(BEL-ANGE@kali)-[~]
$
```

Conclusion

Il était question pour nous dans ce tp de nous mettre dans la peau d'un analyst cybersécurité chargé de répondre à certaines questions.